# FIRSTLOOK

## An Information-Driven Inspection Planner for Post-Storm Power Restoration

### Know where to look next.

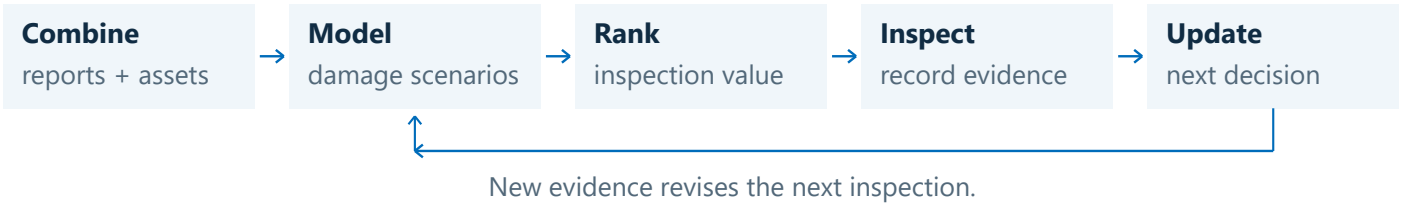
### 01 THE PROBLEM

After a typhoon, outage reports can arrive before the full damage pattern is known. Crews must investigate causes and assess repair needs [1, 2]. With limited assessors, incomplete reports and blocked roads, dispatchers face a consequential choice: **which inspection will most improve the next restoration decision?** A visit that distinguishes a shared supply-line fault from several local failures could clarify the crews and equipment required across multiple communities.

### 02 THE PROPOSED SOLUTION

FIRSTLOOK is a proposed decision-support platform that ranks inspections by their expected information value and time cost. It combines electrical connectivity, asset locations, outage reports and road access into a map of known conditions and plausible damage scenarios, including multiple faults. Each recommendation explains **what the visit could establish, which uncertainty it addresses and why it ranks highly.** Field observations update the scenarios and next recommendation; inconclusive reports preserve uncertainty. Dispatchers retain control of assignments and access restrictions.

### 03 HOW IT WORKS




### 04 TECHNICAL INNOVATION

The integration connects graph-based infrastructure reasoning, sequential evidence updates and explainable routing. The proposed engine will rank expected reduction in scenario uncertainty per travel-plus-inspection minute, subject to access and dispatch priorities. Disclosed scenario weights, uncertain observations and sensitivity checks expose assumptions. **Value comes from how an inspection improves the next decision.** Recommendations remain conditional on scenario coverage and data quality.

### 05 EXPECTED IMPACT

Utility dispatchers could identify damage and repair-resource needs sooner, reducing avoidable assessment travel. Crews could receive clearer inspection objectives; communities could benefit from better-informed recovery planning. Evaluation will assess these potential gains against existing inspection strategies.

### PROTOTYPE SCOPE + EVALUATION

We will build a prototype covering 20 synthetic assets, two assessment crews and multiple damage scenarios, using Python / NetworkX, FastAPI and React / Leaflet. We will compare assessment time, travel and time to identify repair-resource needs against nearest-first and fixed-order inspection baselines. The prototype will use synthetic utility data.

[1] Meralco: Outages and Brownouts

[2] EPRI: Distribution Operations